

# The Agent Loop

Think. Act. Observe. Decide. The four-step cycle that turns a language model into something that can actually get things done.

01

## Chatbot vs. Agent

Ask an AI to reschedule your meeting, check a room, and send invites. A chatbot tells you how. An agent actually does it.

### STANDARD LLM

*"Here's what I'd suggest: open your calendar, find the meeting, drag it to tomorrow..."*

Describes actions

vs

### AI AGENT

*"Done. I moved the meeting to tomorrow at 3pm, booked Room B, and sent updated invites to all 4 attendees."*

Performs actions



#### No Live Data

Anything about current state is stale or fabricated



#### No Actions

Can only describe what to do, not do it



#### No Self-Correction

If wrong, can't check or try a different approach

# The Four-Phase Loop

The agent loop transforms an LLM from a one-shot oracle into an **iterative problem-solver**. Four phases, repeated until the task is complete.



## 1 Think

Examine the current context and reason about what to do next. This is where the model's intelligence lives.

## 2 Act

Execute a concrete action: call an API, query a database, run code, or generate a response.

## 3 Observe

The result comes back — data, an error, a confirmation. New information feeds into the LLM's context.

## 4 Decide

Is the task complete? If yes, return the answer. If not, loop back to Think with the new information.



**Agents can handle ambiguity.** They don't need perfect instructions upfront — they explore, adapt, and try again. This is fundamentally different from traditional software where every branch must be coded in advance.

03

## An Agent in Action

Task: "Find the top 3 bugs by user impact from Sentry, summarize them, and post to #engineering on Slack."

### Loop 1 – Gather

**Think:** *I need to query Sentry for recent issues sorted by frequency.*

```
sentry.listIssues({ sort: "frequency", limit: 10 })
```

10 issues returned with titles and error counts

### Loop 2 – Refine

**Think:** *Frequency ≠ user impact. I need unique users affected for the top 5.*

```
sentry.getIssueStats({ field: "users_affected" })
```

User counts received – rankings change when sorted by impact

### Loop 3 – Deliver

**Think:** *Top 3 identified. Writing summary and posting to Slack.*

```
slack.postMessage({ channel: "#engineering", text: "Bug Impact Report..." })
```

Posted successfully. Task complete.

3 loops · 4 tool calls · 1 output – no step was hard-coded

# Prompt vs. Workflow vs. Agent

Agents are powerful — and slow, expensive, and unpredictable. Use the right tier for the job.

| SIMPLE PROMPT                                                          |      | WORKFLOW                                                                |             | AGENT                                                                                 |               |
|------------------------------------------------------------------------|------|-------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------|---------------|
| Ambiguity                                                              | None | Ambiguity                                                               | Low         | Ambiguity                                                                             | High          |
| External data                                                          | No   | External data                                                           | Yes (known) | External data                                                                         | Yes (dynamic) |
| Speed                                                                  | Fast | Speed                                                                   | Medium      | Speed                                                                                 | Slow          |
| Cost                                                                   | Low  | Cost                                                                    | Medium      | Cost                                                                                  | High          |
| <i>"Summarize this doc" · "Draft an email" · "Translate this text"</i> |      | <i>"Classify ticket → route → draft reply" · "Daily metrics digest"</i> |             | <i>"Why did churn spike?" · "Investigate this login bug" · "Research competitors"</i> |               |

## How much ambiguity is there?

If the task is **predictable**, use a simpler approach. If it requires **exploration and judgment**, that's where agents shine.

### KEY TAKEAWAYS

## What Every PM Should Know

1

**LLMs describe. Agents do.** The gap between a chatbot and an agent is the ability to take action and observe results.

2

**The loop is the secret.** Think → Act → Observe → Decide, repeated until the task is done. No step is hard-coded.

3

**Agents handle ambiguity.** They don't need perfect instructions — they explore, adapt, and self-correct. Traditional software can't do this.

4

**Match the tier to the task.** Prompts for simple, workflows for predictable, agents for open-ended. Using an agent when a workflow will do is like hiring a surgeon to apply bandaids.